

XMBA 200S: Data and Decisions

Chapter 23: Multiple Regression – Part 1
Introduction and Interpretation

Instructor: Reed Walker

Course overview
Part 4: Building regression models

- Blocks 3 and 4
- You want to build a model to forecast outcomes from data you have
- What's the best model you can build?
- This class: models with more than 1 explanatory variable

What's the end goal for today?

- Understanding how to think about models with multiple explanatory factors
  - How do we interpret regression output describing these relationships?
  - How do we interpret estimates when explanatory variables are correlated with one another? Or correlated with other determinants of response variable?

Why is this important?

- Having one explanatory variable is nice, but limiting
- Models with many explanatory variables crucial for making accurate forecasts / predictions

The most important industry in the world

Coffee

Peet's coffee was founded in 1966. The original outlet is still located on the corner of Walnut and Vine in Berkeley

We'll talk about where to locate a new cafe

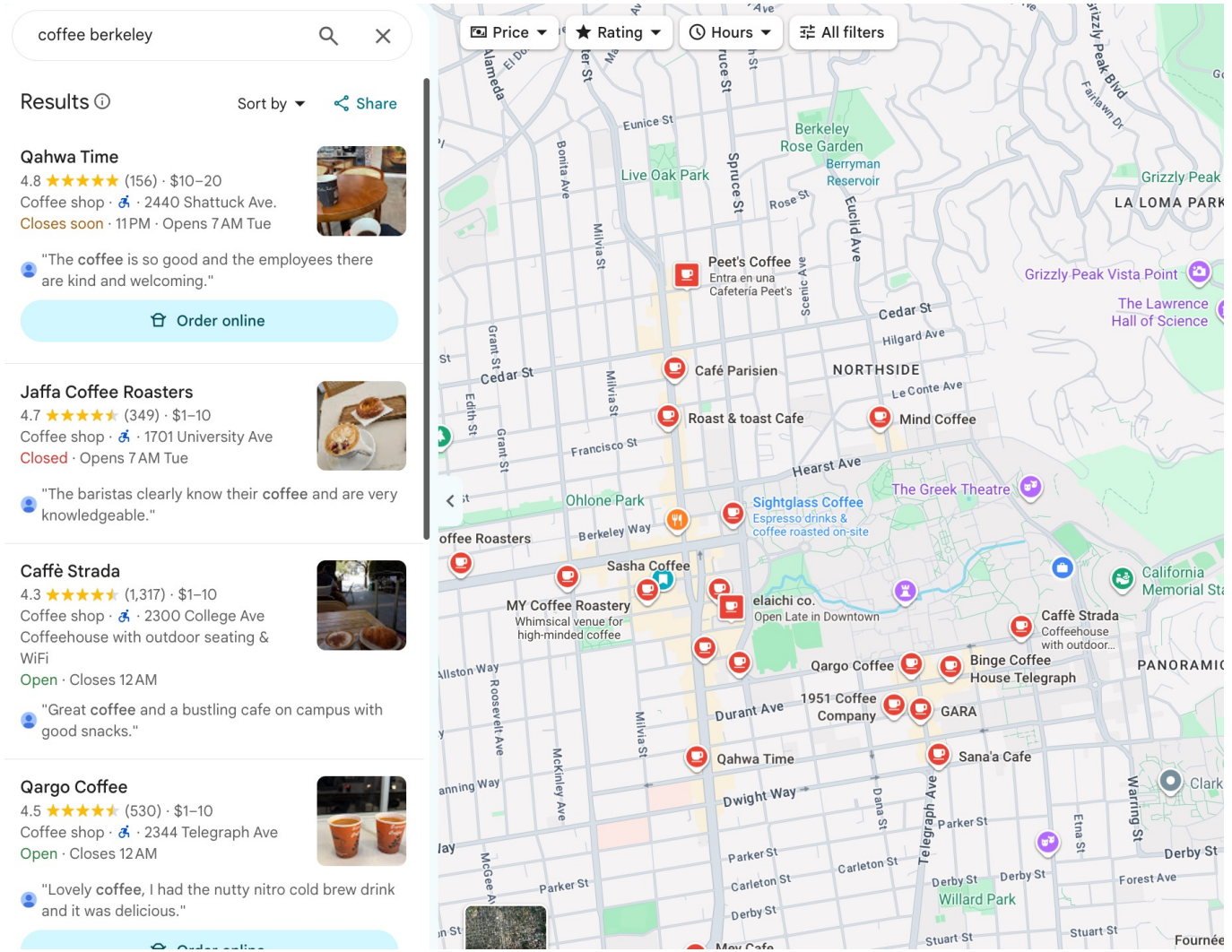






The most important industry in the world

How to think about competition here?



The MRM for the "population"

The point is we now have a bunch of relationships of interest – a bunch of "slopes"

Observed values of the response y are linearly related to the values of the explanatory variables $x_1, x_2, \ldots, x_k$ by the equation:

$$y = \beta_0 + \beta_1 x_1 + \underbrace{\beta_2 x_2 + \cdots + \beta_k x_k}_{\text{new relative to SRM}} + \varepsilon, \qquad \varepsilon \sim N(0, {\sigma_\varepsilon}^2)$$

Note: No one except the book calls these SRM or MRM. They are all referred to as a "regression model"

Where to locate a coffee shop?

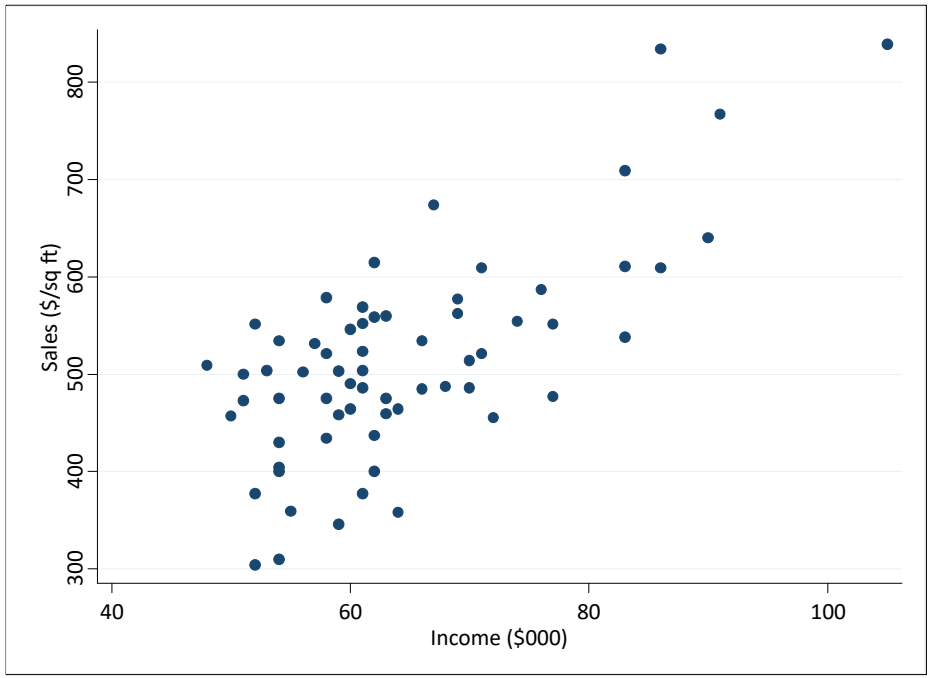
- You have data from a sample of 65 Peet's retail locations.
- Based on these data, where should Peet's locate new stores?
- What is the relationship between sales, competition, and local income?
- Which factor is more important? Is it better to be in a high-income area with lots of competitors, or in a lower-income area with less competition?

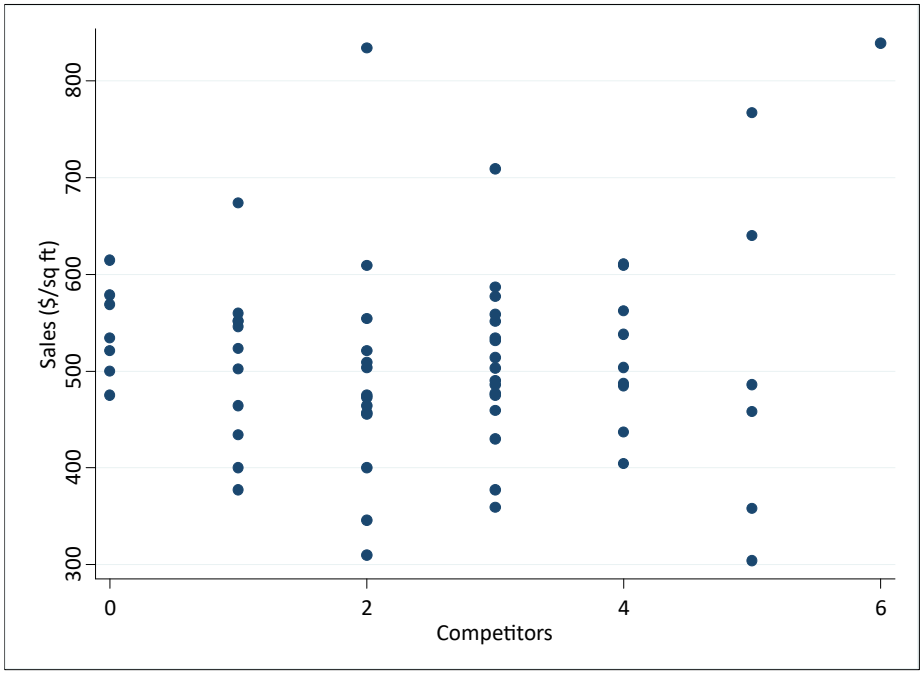
Where to locate a coffee shop?

What goes into our model?

- Response variable: annual sales at Peet's outlets per square foot of retail space
- Two explanatory variables:
  - Median household income in the area (thousands of dollars)
  - Number of competing stores within ½ square mile

Look at your sample data: Scatterplots





Response variable (y-axis):
Annual sales at Peet's outlets per square foot of retail space

Interpreting Multiple Regression

Suppose you regress sales on local income.

Regression Statistics	
R Square	.5013
Adjusted R Square	.4933
Standard Error	74.869
Observations	65

	Coefficients	*Standard Error*	*t-Stat*	*P-value*
Intercept	97.07	53.18	1.83	0.073
Income ($, in 1000s)	6.46	0.81	7.96	0.000

Interpreting Multiple Regression

Suppose you regress sales on the number of competitors.

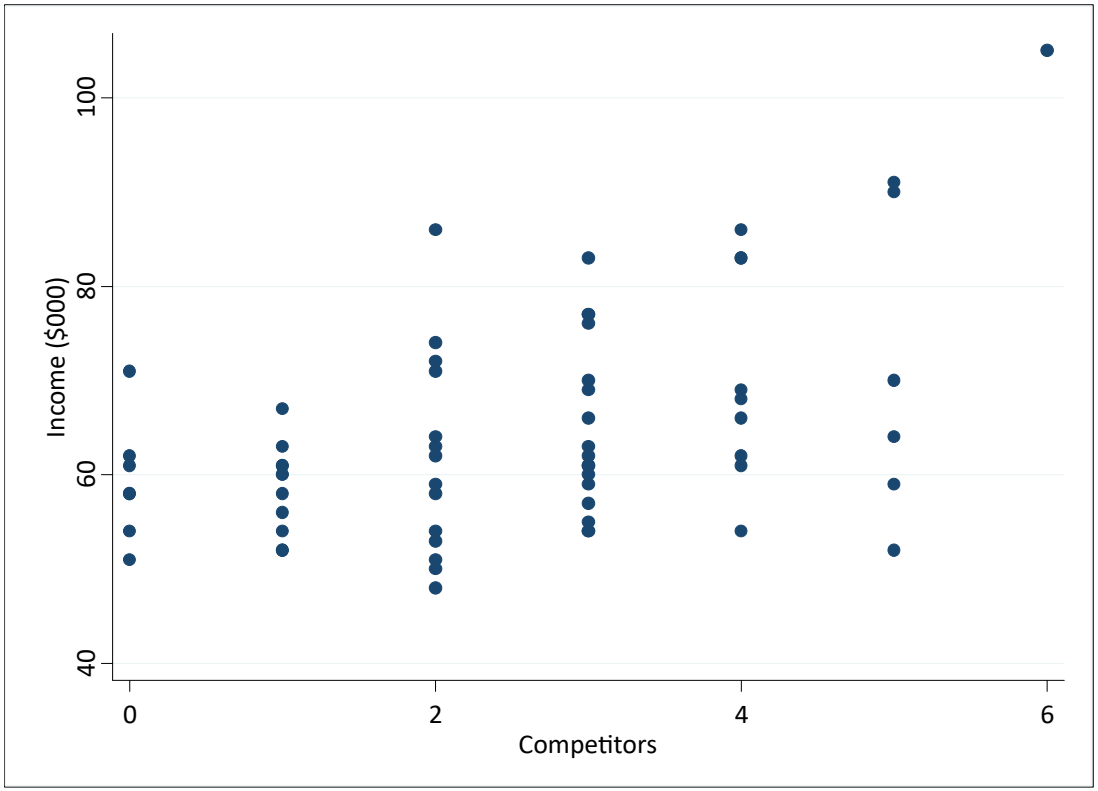
Regression Statistics	
R Square	.0044
Adjusted R Square	-.0114
Standard Error	105.78
Observations	65

	Coefficients	*Standard Error*	*t-Stat*	*P-value*
Intercept	502.2	25.44	19.74	0.000
Competitors	4.64	8.75	0.53	0.598

Interpreting Multiple Regression

Another Scatterplot and Correlation Matrix

- Positive correlation between income and competitors



	Sales	Income	Competitors
Sales	1.0000	0.7080	0.0666
Income	0.7080	1.0000	0.4743
Competitors	0.0666	0.4743	1.0000

Interpreting Multiple Regression

Peet's Regression Results

R^2	0.5947
$\overline{R}^2$	0.5816
s_e	68.0306
n	65

Term		Estimate	Std Error	*t*-Statistic	*p*-Value
Intercept	b_0	60.359	49.290	1.22	0.2254
Income	b_1	7.966	0.838	9.50	$<.0001$
Competitors	b_2	-24.165	6.390	-3.78	0.0004

Estimated Sales= 60.36 + (7.97) *Income* – (24.17) *Competitors*

Interpreting Multiple Regression Estimates

- The slope b_1 = 7.966 for *Income* implies that…
  - a store in a location with $10,000 more in median household sells, on average, $79.66 more per square foot than a store in a less affluent location with the same number of competitors
- The slope b_2 = -24.165 for *Competitors* implies that…
  - among stores in equally affluent locations, each additional competitor lowers average sales by $24.17 per square foot
- In all cases, "what is the association between the response variable and the explanatory variable, **holding all other explanatory variables fixed**"

Multiple Regression: Thought Exercise

How does this work in practice?
Analogy: multiple regression as a matching exercise

- Imagine if you could observe:

1. A large number of coffee shops with the same income, but a different numbers of competitors

AND

2. A large number of coffee shops with the same number of competitors, in neighborhoods with different incomes

Then you could isolate the effect of just one variable, “holding the other variable `fixed’”

“Holding income fixed, the relationship between sales and competitors is negative and statistically significant…”

Marginal vs. Partial Slopes (from book: ignore)

Partial slope: slope of an explanatory variable in a multiple regression that statistically excludes or "controls for" the effects of other explanatory variables

~~**Marginal slope:** slope of an explanatory variable in a simple regression~~

- ~~Marginal slope estimate may combine direct effect with other channels of influence (i.e. bias from lurking variables)~~

~~*Partial and marginal slopes are the same only when the explanatory variables are uncorrelated.*~~

The book discusses "marginal slopes", but this is not a term that is used very often in statistics

Measures of explanatory power

- R^2 measures the fraction of variation in *y* explained by the explanatory variables
  - In the coffee example, 59% of the store-to-store variation in sales is explained

Typically the focus of most analysts

- The adjusted R^2 adjusts for the sample size *n* and model size *k* (where *k* is the number of explanatory variables)
  - It is always smaller than R^2. $\bar{R}^2 = 1 - (1 - R^2)\left(\frac{n-1}{n-k-1}\right)$.

- The standard deviation of the residual s_e is exactly the same as before
  - And, you can use this to estimate prediction intervals, as you did with simple regression

Inference in Multiple Regression

Prediction Intervals

As in the SRM, an approximate 95% prediction interval is given by,

$$\hat{y} \pm 2\, s_e.$$

We again need to be careful not to make prediction intervals outside of the range for which we have data.

Inference in Multiple Regression

Prediction Intervals for Peet's

So, for example, predicted sales per square foot at a location with household income of $70,000 and 3 competitors is,

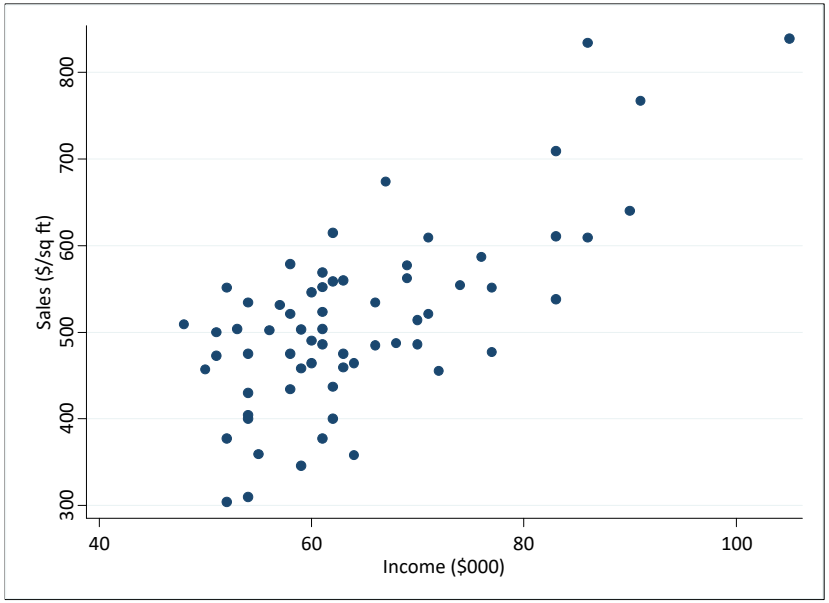
$$60.36 + (7.97)\,Income - (24.17)\,Competitors$$
$$= 60.36 + (7.97)(70) - (24.17)(3) = \$545$$

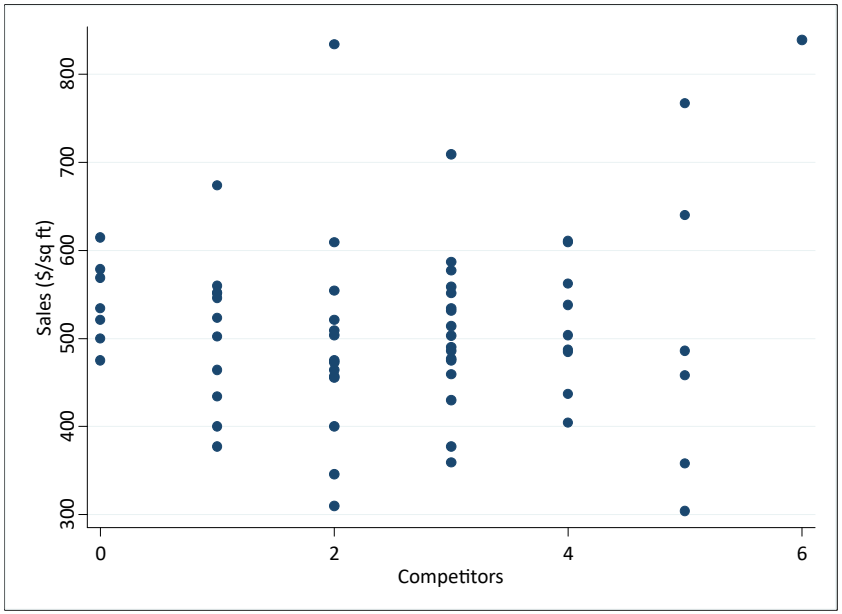
So, the 95% prediction interval for predicted sales per square foot is approximately,

$$\$545 \pm \$136 \text{ or } [\$409, \$681].$$

Nonlinear Patterns in MRM

In the Peet's coffee example the relationships were approximately linear, so we did not consider taking logs.





But would it make sense to put sales in logs?

Semi-Log Models

One alternative would be to estimate a "semi-log" model,

$$\log(Sales) = b_0 + b_1 Income + b_2 Competitors$$

Regression Statistics	
R Square	.5532
Adjusted R Square	.5388
Standard Error	0.137
Observations	65

	Coefficients	*Standard Error*	*t-Stat*	*P-value*
Intercept	5.40	0.099	54.47	0.000
Income	0.015	0.002	8.76	0.000
Competitors	-0.051	0.013	-3.98	0.000

- A $1000 increase in neighborhood income is associated with a 1.5 percent increase in sales per sqft, holding competition fixed
- A one unit increase in competitors associated with 5 percent reduction in sales per square feet, holding income fixed

Mixed-Log Model

Or we could even estimate a model with a mix,

$$\log(Sales) = b_0 + b_1 \log(Income) + b_2 Competitors$$

Regression Statistics	
R Square	.5221
Adjusted R Square	.5067
Standard Error	0.141
Observations	65

	Coefficients	*Standard Error*	*t-Stat*	*P-value*
Intercept	2.26	0.482	4.69	0.000
Log (Income)	0.982	0.119	8.23	0.000
Competitors	-0.047	0.013	-3.60	0.001

- A 1 percent increase in neighborhood income is associated with a 0.982 percent increase in sales per sqft (i.e. elasticity), holding competition fixed
- A one unit increase in competitors associated with 5 percent reduction in sales per square feet, holding income fixed

Best Practices and Pitfalls

Almost everything is the same with the MRM.

Difference is interpretation: “holding other variables fixed”

Useful to examine scatterplots and the correlation matrix before running regressions.

Avoid causal interpretations! There will always be lurking variables and confounding (unless this is an experiment)

Big Picture Recap + Takeaways

- This class: The mechanics of reading regression output in MRM:
  - You need to understand “ceteris paribus”.

- More than that, this is a *way of thinking*:
  - Correlations hide multiple channels of association.
  - Unless you account for all of these you won’t get the “pure” or “causal” effect of a particular variable. Instead, you get a mixture of channels.
  - If you don't have data on one of the channels, it will be embedded in what you do control for / estimate (i.e. a lurking variable/confounding), and you need to take that into account.

- Bottom line: Thinking about confounding variables and direct + indirect channels is what allows you to make sense of the data you observe.
  - If you think the channels you don’t observe aren’t too important, then you’re fine.
  - If you think they are important, then think about how they distort your inferences and your decision making.

Administrative details: the final exam

- The final exam will be on Friday 11/13, 1-4pm.
- The final exam is an individual assignment:
  - Open book / notes
  - Computers, tablets, and cell phones not allowed

Administrative details: the final exam

What to expect?

- A cumulative exam, but with emphasis on later material
- More emphasis on concepts/applications/interpretations (e.g., team exercise questions)…
- Previous assignments useful proxy for types of questions
  - interpreting regression output,
  - design and analyze experiment,
  - best practices for experimental design/implementation,
  - challenges/issues in data collection,
  - bias versus precision, how to use p-values, etc...

Preparing for the final

- Solve Problems!
  - Knowledge Check Problems, Team Assignments
- Go through the slides, review videos, review materials posted on bCourses
- Opportunities to talk with me and Silvia:
  - Office hours
  - Review session(s)
  - Email to schedule an appointment!